Gondwana Research 51 (2017) 177–192

Contents lists available at ScienceDirect

Gondwana Research

journal homepage: www.elsevier.com/locate/gr

Magnetostratigraphy across the Triassic-Jurassic boundary in the main Karoo Basin

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ARTICLE INFO

Article history:
Received 7 October 2016
Received in revised form 26 June 2017
Accepted 24 July 2017
Available online 29 July 2017

Handling Editor: J.G. Meert

Keywords:
Magnetic stratigraphy
Elliot Formation
Late Triassic
Early Jurassic
Karoo Basin

ABSTRACT

The end-Triassic mass extinction and the transition and explosive diversification of fauna over the Triassic-Jurassic boundary is poorly understood and poorly represented in the rock record of the Southern Hemisphere. This is despite the rich diversity in both body and trace fossils of Triassic-Jurassic age in southern Africa, which is not found in coeval Northern Hemisphere localities. We report here the first palaeomagnetic polarity zonation of the Upper Triassic-Lower Jurassic continental red bed succession (Elliot Formation; Stormberg Group) in southern Africa. The results from 10 partially overlapping sections, with a composite thickness of ~280 m, provide a magnetic polarity chronology of the main Karoo Basin in South Africa and Lesotho. Palaeomagnetic analyses reveal that heating samples to between 150 °C and ~300 °C removes the secondary, moderately inclined (~48°) normal-polarity component of remanent magnetization. This component overlaps with the present-day field and is comparable to the overprint direction expected from Lower Jurassic Karoo dolerite intrusions. In contrast, a likely primary, high unblocking temperature component, of dual polarity, consistently is of steeper inclination (~63°). This characteristic remanence passes the reversals test, except where means are based on small sample populations. There are only two resulting polarity zones for the ~200 m thick lower Elliot Formation (LEF) with potential for a thin 3rd magnetozone in the uppermost part. The upper Elliot Formation (UEF), in contrast, which was sampled over a thickness of ~ 80 m, has five polarity zones. The failure of the reversal test for the UEF and combined Elliot Formation (LEF + UEF) indicates that the normal polarity samples may be biased by a younger overprint of either the Jurassic normal polarity of the Karoo Large Igneous Province or present day field. The separate poles calculated for the four sites in the LEF and ten sites in the UEF overlap with the Late Triassic and Early to Middle Jurassic Gondwana poles, respectively. The combined Elliot Formation and UEF pole positions are better constrained than the LEF and therefore considered more reliable. Overall the LEF shows considerable overlap with the Late Triassic Apparent Polar Wander Paths (APWP) poles.

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1. Introduction

The stratigraphy of continental deposits of the Late Triassic–Early Jurassic has been globally investigated via magnetostratigraphy, biostratigraphy and geochronology (radiometric dating, astrochronology - e.g., Olsen et al., 2011; Blackburn et al., 2013). An important 'backbone' onto which many continental and marine successions of this age are compared is provided by the Newark Basin (USA) and Hartford Basin (Canada) composite astrochronostratigraphic polarity time scale (APTS). Kent et al. (2014b) currently consider the Newark-Hartford APTS ages to be coeval with radiometric dates in marine successions for the Norian-Rhaetian boundary (Wotzlaw et al., 2014), the Triassic–Jurassic boundary (TJB; Guex et al., 2012) and the Hettangian–Sinemurian (Schaltegger et al., 2008).

The end-Triassic event (ETE; 201.6 Ma), one of the 'big five' Phanerozoic mass extinction events, has been linked to the onset of the Central Atlantic Magmatic Province (CAMP; Blackburn et al., 2013). Understanding of the ETE and its causal link to the CAMP is largely based on evidence from the Northern Hemisphere (principally North America, Europe and Morocco). To date, there is a limited number of studies that present evidence for the ETE from a Southern Hemisphere perspective. This is a function of the number of fossiliferous and accessible localities bearing datable Upper Triassic-Lower Jurassic sedimentary rocks.

In South Africa and Lesotho, the ETE and the TJB have been considered to be represented in the continental red bed succession of the richly fossiliferous Elliot Formation (Stormberg Group, Karoo Supergroup) that spans the Late Triassic and Early Jurassic (Olsen and Galton, 1984). However, no formal radiometric ages or palaeomagnetic data

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http://dx.doi.org/10.1016/j.gr.2017.07.009
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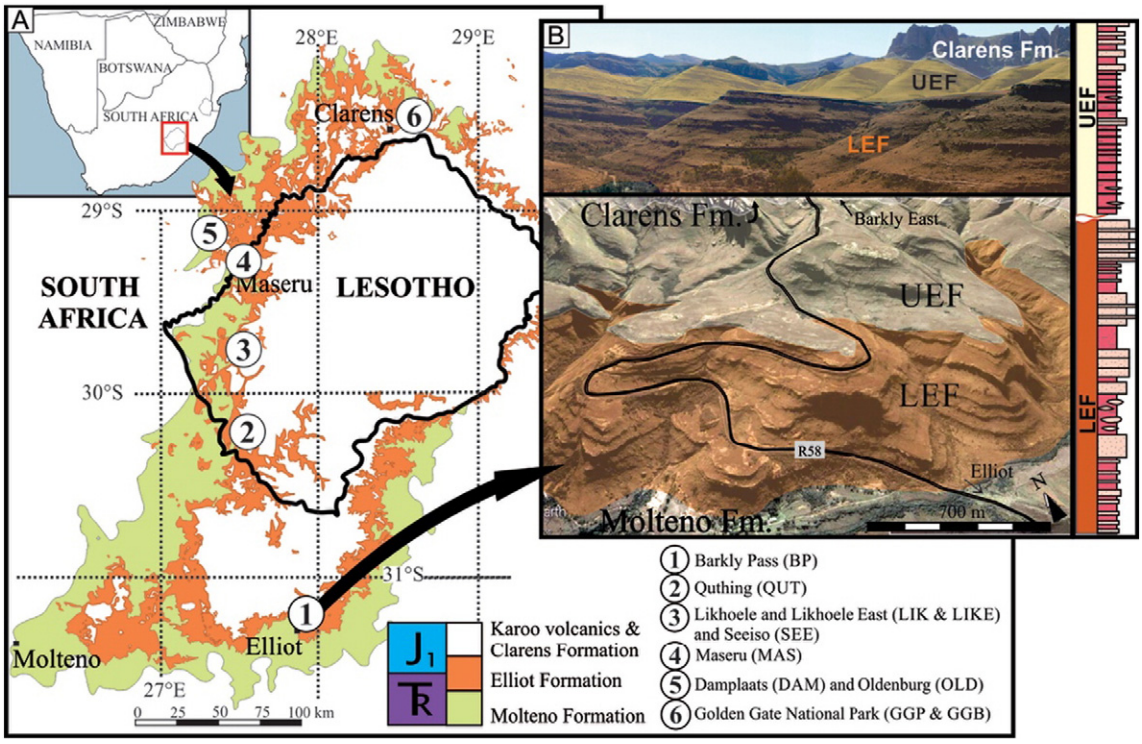


Fig. 1. (A) Simplified geological map of the Elliot Formation (Stormberg Group, Karoo Supergroup) and the study areas (numbered 1–6) in South Africa and Lesotho; (B) Holostratotype at Barkly Pass (section #1) illustrated in a field photograph, generalised geological map (modified from a Google Earth image) and log with the lower (LEF) and upper (UEF) Elliot Formation highlighted in orange and light yellow, respectively. Note that the continental flood basalts of the Karoo Large Igneous Province (KLIP) are conformably underlain by the Clarens Formation, which in turn has a conformable lower contact with the Elliot Formation.

have been published for the Elliot Formation, and to date, this age assignment depends solely on its poorly defined vertebrate biostratigraphy and stratigraphic position below the well-dated Karoo Large Igneous Province (KLIP; Fig. 1). We present here the first magnetostratigraphic data collected for the Elliot Formation across the main Karoo Basin at localities in South Africa and Lesotho.

In this study, we present a global magnetostratigraphic correlation of the Elliot Formation. The richly fossiliferous Elliot Formation has the potential to shed light onto the rise of the dinosaurs and early mammaliaforms as well as the post-extinction environmental recovery in the wake of the ETE. This study provides a foundation onto which further stratigraphic investigations can be built, including, but not limited to, the (a) study of faunal turnover within the earliest Jurassic in Gondwana and (b) the assessment of the timing of dinosaur dispersal.

1.1. Geologic and stratigraphic background

The lithostratigraphy of the Elliot Formation is summarized in Bordy and Eriksson (2015). Currently, the formation is informally subdivided into two subunits, the lower Elliot (LEF) and upper Elliot (UEF) formations, based on local and regional scale differences in sediment provenance, palaeocurrent patterns, petrography, sedimentary structures and geometry of sandstone bodies (Bordy et al., 2004a, 2004b, 2004c). These differences in the sedimentary facies suggest a stratigraphic gap of unknown duration at the contact of the LEF and UEF. The sedimentary and biofacies differences in the underlying Carnian Molteno Formation imply that the lower contact of Elliot Formation is also unconformable (Bordy et al., 2005). The upper contact of the Elliot Formation is conformable as suggested by the regional intercalations with the overlying mainly aeolian to fluvio-lacustrine Clarens Formation (Bordy et al., 2004a). The thickness of the Formation ranges from ~70 m in the north to ~460–480 m in the south of the basin (Bordy et al., 2004a). The holostratotype is in the Barkly Pass (Fig. 1), where the formation is exposed in near entirety. Tilting of the strata is negligible (≤5°) throughout the basin.

To date, an unpublished MSc dissertation by De Kock (2003) presents the only magnetostratigraphic study of the Elliot Formation. A palaeopole from mostly the UEF (68.9°N, 265.6°E; De Kock, 2003) was found to compare well with the mean Early Jurassic African palaeopole of McElhinny and McFadden (2000).

The overlying Drakensberg Group is a series of Lower Jurassic volcanic and subvolcanic units (chiefly dolerite dykes and sills) that are representative of the KLIP, and which ended the sedimentation within the main Karoo Basin. Van Zijl et al. (1962a, 1962b) first undertook a palaeomagnetic study of the continental flood basalts of Drakensberg Group in Lesotho. A sampled section within the Maseru District provided the magnetic polarity of several samples within the Clarens Formation sandstones (3 unbaked and 6 baked) directly below the basalts. This study found that the lowermost basalts have a reverse magnetic polarity, with the basalts higher up in sequence being normally magnetized. Later palaeomagnetic studies have repeatedly shown this single reversal ('Van Zijl' reversal; Kosterov and Perrin, 1996; Hargraves et al., 1997; Prévot et al., 2003; Moulin et al., 2011, 2012, 2017), which in conjunction with geochemical data (Marsh et al., 1997) suggest a very rapid outpouring of the Karoo lavas in either 3000 yr or 4 myr timespan (Moulin et al., 2011; Jourdan et al., 2008, respectively). Ultimately, all magmatism from the northern Free State and KwaZulu Natal provinces (South Africa), and Antarctica (Kirwan Mountains) fall into the range 183 ± 1.0 Ma (Jourdan et al., 2007). Given the intercalation of the uppermost Clarens Formation with the lowermost lava flows, it is conservative to assume that at least the topmost section of the Clarens Formation is contemporaneous with the lowermost KLIP age.

Palaeontologically, the Elliot Formation is notable for its vertebrate and ichnofossil assemblages (Ellenberger, 1970; Kitching and Raath, 1984; Knoll, 2004, 2005). The current informal lithostratigraphic units

correspond to the biostratigraphic zones of Kitching and Raath (1984; Fig. 1), with the LEF containing the '*Euskelosaurus*' Range Zone, and the UEF comprising the *Massospondylus* Range Zone (Bordy et al., 2004a). The contact between the '*Euskelosaurus*' and *Massospondylus* range zones has been traditionally considered to represent the TJB (Smith and Kitching, 1997). In terms of Late Triassic sauropodomorph fauna, the LEF is one of the richest successions in Southern Africa and Gondwana (Yates, 2007a, 2007b; McPhee et al., 2016). Fossil assemblages of the LEF have been considered to be Norian in age (Knoll, 2004; Lucas and Hancox, 2001), and the sauropodomorphs, at least, are considered contemporaneous with those of middle Norian in Germany, the Norian Los Colorados Formation in the Ischigualasto–Villa Union Basin of Argentina and the uppermost Chinle Formation of North America (Yates and Kitching, 2003; Yates, 2007a, 2007b). The faunal assemblages of the UEF are conservatively considered earliest Jurassic, more specifically Hettangian to Sinemurian, with an age range of ~200 to <190 Ma (Lucas and Hancox, 2001). Norman et al. (2011) speculated that the age of the UEF may be younger and either spans the Pliensbachian-Toarcian or the upper Sinemurian based on comparative work by Yates et al. (2004) and personal observations presented by Irmis (2010).

The age of the Elliot Formation thus has long and almost solely rested on the presence of globally comparable fossil material (e.g., Du Toit, 1954; Romer, 1970; also see Kitching and Raath, 1984; Yates and Kitching, 2003). However, because the ages of different continental faunal assemblages traditionally rests on global comparative studies, there have been numerous revisions of the relative ages of these assemblages as more numeric dates become available globally (Sues, 2016). To date, the diverse and abundant body and trace fossils assemblages of the Elliot Formation suggest an age range from uppermost Late Triassic (Norian/Rhaetian) to lowermost Early Jurassic (~Sinemurian/Pliensbachian) (e.g., Kitching and Raath, 1984; Olsen and Galton, 1984; Lucas and Hancox, 2001; Knoll, 2004, 2005; Norman et al., 2011). The palaeontologically established age of the Elliot Formation is therefore in agreement with the radiometric dating of the overlying Toarcian flood basalts of the Drakensberg Group (~183 Ma; Duncan et al., 1997; Jourdan et al., 2007). Thus far, there are no published radiometric ages for the Elliot Formation; however, there are references to tuffaceous strata in Lesotho (Rooyani and Schmitz, 1987) and bentonite in South Africa (Botha and Theron, 1967; Johnson, 1971; Bordy and Abrahams, 2016).

2. Materials and method

All palaeomagnetic samples were collected by drilling orientated cores using a portable drill and a sun compass. All Elliot Formation rock types were sampled, including mud-chip and pedogenic nodule conglomerates to sandstones, siltstones, mudstones and palaeosol horizons. Herein, we define sampling sections as the stratigraphic sections from which several palaeomagnetic samples were obtained. Wherever the lithology allowed, sampling resolution was relatively high (av. 1 sample/~50 cm). This was in order to increase the resolution of the study, to obtain the true representation of the change in the magnetic field during the deposition of the Elliot Formation and to monitor any homogeneity in the magnetization between and within sections. To lessen the bias from the stratigraphic incompleteness (i.e., nondeposition, gaps in the sequence etc.; Miall, 2014b), the density of sampling was adjusted to sedimentary facies distributions (i.e. pedogenically altered mudstones were sampled more densely than unaltered ones). Sections drilled were chosen for the relative ease of access, the fossil content, previous work (e.g., palaeontological, geochemical) and the absence of conspicuous Lower Jurassic Karoo dolerite intrusions. Dolerite intrusions can magnetically overprint the sedimentary rocks, giving a secondary Early Jurassic partial (or complete) remagnetization in the vicinity of the intrusions. Completely avoiding igneous intrusions is difficult because in some areas the intrusions are very pervasive and the country rock alteration is prominent. Furthermore, the extent of the alteration is highly variable between intrusions, where other factors may come into play such as fracturing, size of intrusion or duration in which intrusion acted as a feeder system. At sections close to the contact of the Elliot-Clarens formations, the Karoo basalts are another source of bias, but this only affected 4 out of the 10 measured sections. Due care was taken to understand and manage any overprint by Karoo igneous bodies.

Demagnetisation data were obtained using principle component analysis (Kirschvink, 1980), including both line and plane analyses. Analyses were performed in the computer programs PaleoMac 5.4 (Cogné, 2003) and Paleomag 3.1 (Jones, 2002). Only demagnetisation lines with maximum angular deviation (MAD) angle values ≤10°, and planes ≤15° were included in the subsequent analysis. For the purpose of polarity stratigraphy, demagnetisation lines with MAD values ≤15° and planes ≤15° were accepted.

The studied stratigraphic sections are located in the northern (2 sections), central (6 sections) and southern (2 sections) parts of the outcrop area of the Elliot Formation in South Africa and Lesotho along a ~350 km transect (Fig.1). As there are no recognisable event markers, to date, within the Elliot Formation, the sedimentological change that defines the lithostratigraphic boundary between LEF and UEF was used as a first order datum for correlations between the sampled sections. The sedimentological break (described in Bordy et al., 2004a) is expressed in the field as a geomorphological plateau. This plateau can be used a tool of correlation because it is a practical way to visually subdivide the formation on a large-scale (Fig. 1B) and, most importantly, is an independent verification tool for stratigraphic correlation. Bed-by-bed correlation between the isolated sampling sections was not attempted because depositional conditions, including sedimentation and preservation rates, are assumed to have been highly variable in this fluvio-lacustrine system (cf. Miall, 2014a, 2014b, 2015). In particular, the high resolution correlation of the sections is limited by the perceived variability and unpredictable nature of the syn-sedimentary erosional events, which most probably resulted in sedimentation gaps with unquantifiable magnitudes and non-uniform spatiotemporal distribution in the Elliot Formation. Such largely cryptic stratigraphic gaps (i.e., frequent paraconformities) are inferred from the high abundance of pedogenic alternation features, reworked pedogenic nodule conglomerates, major thickness differences along the N-S axis of the basin and the overall long timespan represented by the Formation (>20 Ma) (Smith and Kitching, 1997; Bordy et al., 2004a, 2004b; Bordy and Eriksson, 2015).

3. Results

Natural remanent magnetization (NRM) was studied by stepwise alternating field and thermal demagnetisation methods. Samples from all sections, with the exception of Oldenburg, show the unblocking of both normal and reverse polarity high temperature magnetization after the removal of a poorly-defined, low temperature (lost by 150 °C and infrequently remaining up to 350 °C; Fig. 2) remanence with inconsistent declination and inclination. Normal polarity samples show a consistent isolation of a remanence with moderately negative, and steeper than the present field, inclination and north-north-westerly declination. Reverse magnetisation showed consistent remanence with a moderate positive inclination and a south-south easterly declination (see Supplementary material). The higher temperature magnetization was interpreted as a characteristic remanent magnetization (ChRM). Fine-grained samples (massive mudstones coded as facies Fm in the logs) appear to preserve the most stable NRM and ChRM in comparison to fine- to medium-grained sandstones and mud-chip conglomerates (facies Sm or Gcm/Gmm; Fig. 2). Previously conducted isothermal remanent magnetization (IRM) analysis (De Kock, 2003) suggests that the high coercivity element for the ChRM is hematite, but that partial carriers may also be maghemite and magnetite in red mudstones.

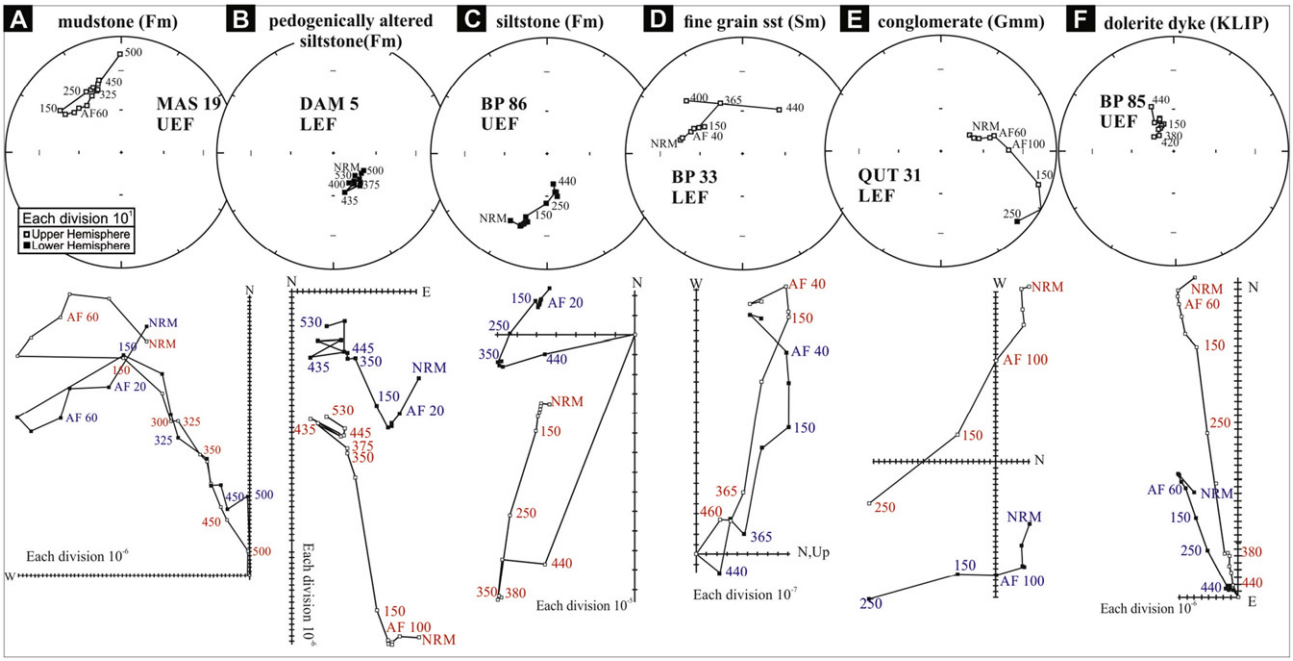


Fig. 2. (A)–(E) Different lithologies as representative examples of equal area projections and orthogonal vector plots from various sections of the Elliot Formation. (F) Sample BP85 is a dolerite dyke which was sampled as a control and illustrates stable direction. Section codes: MAS = Maseru, DAM = Damplaats, BP = Barkley Pass, QUT = Quthing, KLIP = Karoo large igneous province.

Palaeomagnetic directions were determined by least-squares analysis (Kirschvink, 1980). Most samples had the following two, sometimes three, well-defined components:

1. Low-coercivity (soft) magnetic component (NRM – 10/15 mT) (coded 'SFT').
2. Present local Earth field component (10 mT/150–350 °C) (generally NNW and upward directed component) (PF).
3. (i) High-temperature steep (i.e., steeper than PF) NW component (between 150 and 350 °C to 440–660 °C) with negative inclination (normal, N), or (ii) High-temperature steep SE component (between 150 and 350 °C to 440–660 °C) with positive inclination (reverse, R; Fig. 2).

Randomly distributed remanence directions that were removed in all samples during low-field AF-treatment, at demagnetization field-strength levels below 10 mT, are considered to be low-coercivity components (soft). These were acquired either in the field through natural processes or through sample preparation. Several samples retained scattered north, north-westerly and upward magnetic component at higher levels of demagnetization. This component is identical to the present local Earth field (PF). Generally consistent in direction, the PF component was often removed by 150 °C, but infrequently remained in some samples up to 350 °C. However, in some samples it is persistent and stable to higher thermal demagnetization steps (e.g., Barkly Pass samples BP1, BP39, Likhoele sample LIK 27 with PF stable to 650 °C). Samples which have been completely overprinted by the present field were excluded from further analysis. Characteristic high-temperature remanence components (N or R) were stable at higher demagnetization levels, up to 580 °C with some samples stable to ~620 °C. For example, Barkly Pass sample BP 73 (UEF) has a soft component present until 350 °C before the unblocking of a normal direction is observed, with higher temperature components (365–620 °C) shifting westwards along a short great circle path, at steeper inclination, and having a north-westerly-directed stable end point vector direction. Most samples displayed this behaviour during increasing levels of thermal demagnetization, and this is considered to represent the normal characteristic magnetization (ChRM) direction. Stable end-point vector directions to the south east were reached along great circle paths for reverse direction samples. For instance, Likhoele sample LIK29 shows a strong present field component (northeast and up-directed) in addition to a stable (up to 625 °C) R-component that demagnetizes along a great circle path. For palaeomagnetic stability and to establish if the LEF and UEF samples have one true mean for reversed and normal samples, the comparison of the antipode of the mean of the reversed-polarity samples is taken against the mean of the normal polarity samples, in a reversals test (see Supplementary material; McElhinny and McFadden, 2000). It was based on the positive overall reversals tests that the ChRM N- and R-components are interpreted to be primary Late Triassic-Early Jurassic directions for the Elliot Formation in addition to the regionally consistent magnetic polarity patterns. Stable end-point vector directions to the south-south-east were reached along great circle paths.

Combined palaeomagnetic data from the 10 sections (with $\alpha 95 < 16°$) covering the LEF-UEF transition results in a mean direction for normal- and reverse-polarity samples for the LEF of Dec = 320.8° and Inc. = −62.6° ($\alpha 95 = 10.0°$, k = 85.09, n = 4 sections), the UEF of Dec. = 327.7° and Inc. = −62.7° ($\alpha 95 = 26.9°$, k = 4.3, n = 10 sections), and Elliot Formation combined (LEF + UEF = EF) with Dec. = 328.6°and Inc. = −64.3° ($\alpha 95 = 26.5°$, k = 4.3, n = 10 sections).

3.1. Late Triassic-Early Jurassic palaeopole of the Elliot Formation

The section average declination, inclination and associated α_{95} for the LEF (N + R), UEF (N + R) and combined Elliot Formation (EF, N + R) in conjunction with the section location (Lat. N, Long. E) were used to calculate the virtual geomagnetic poles (VGPs) for the various sections, in addition to the mean palaeomagnetic poles (Table 1).

Twenty-four new VGPs, of various qualities, were calculated for the LEF, UEF and the Elliot Formation. Age constraints for these poles are taken from the biostratigraphic evidence (e.g., Lucas and Hancox, 2001; Knoll, 2004, 2005; Irmis, 2010; Norman et al., 2011), which considers the LEF to be Late Triassic (~213–~201 Ma), and the UEF to be Early Jurassic (~201–~195 Ma), with an approximate age of the Elliot Formation ranging from ~213 Ma (Norian) to ~195 Ma (Sinemurian).

Table 1
Elliot Formation sections and their section means in conjunction with the co-ordinates of the present day section location, the calculated pole location and the palaeopole. Mean virtual geomagnetic poles are given for the combined Elliot Formation (EF), lower Elliot Formation (LEF) and upper Elliot Formation (UEF). α95 = 95% confidence interval, pole; A95 95% confidence interval; n = samples per section, N = number of sections.

Elliot Formation combined (LEF + UEF)					Section location		Pole location				VGP	
Section #	n	Decl °	Incl °	α95 °	Lat °N	Long °E	Pole lat °N	Pole long °E	dm	dp	Lat °N	Long °E
BP EF	62	303.2	−70.7	4.4	−31.4	27.3	44	249	7.6	6.6	−44.0	69.2
QUT EF	64	339.8	−59.8	4.4	−30.4	27.6	70.7	260.0	6.6	5.0	−70.7	80.0
MAS EF	24	332.5	−51.8	9.5	−29.3	27.5	66.3	282.9	13.0	8.8	−66.3	102.9
DAM EF	45	314.5	−61.1	5.5	−29.2	27.3	51.4	265.1	8.4	6.5	−51.4	85.1
GGP EF	36	319.1	−56.8	6.2	−28.5	28.7	54.8	273.3	9.0	6.5	−54.8	93.2
GGB UEF	14	345.9	−55.8	15.5	−28.5	28.7	75.8	261.7	22.2	15.9	−75.8	81.7
LIK EF	26	5.7	−68	8	−29.3	27.5	67.8	198.0	13.4	11.3	−67.8	18.0
LIKE EF	15	306.2	−66	10.4	−29.9	27.3	45.5	257.2	17.0	13.9	−45.5	77.2
OLD UEF	12	317.7	−61.2	12.4	−29.1	27.4	53.6	264.5	19.0	14.6	−53.6	84.5
SEE UEF	23	143.1	46.9	8.4	−29.8	27.5	−57.8	111.2	10.8	7.0	−57.8	111.2
		EF: Elliot Formation combined (LEF + UEF)						Decl °	Incl °	α95 °	k	
							Sect. mean	328.6	−64.3	26.5	4.3	
								Lat °N	Long °E	A95 °	K	
							Mean pole	−60.47	082.06	9.5	27.04	N = 10

Lower Elliot Formation (LEF)					Section location		Pole location				VGP	
Section #	n	Decl °	Incl °	α95 °	Lat °N	Long °E	Pole lat °N	Pole long °E	dm	dp	Lat °N	Long °E
BP LEF	16	312.6	−66.7	8.4	−31.4	27.3	51	256	13.9	11.4	−50.5	76.4
QUT LEF	36	338.9	−59.1	6.5	−30.4	27.6	70.4	263.0	9.7	7.3	−70.4	83.0
DAM LEF	38	303.8	−66.3	13.6	−29.2	27.3	43.4	256.3	22.3	18.3	−43.4	76.3
GGP LEF	20	322.2	−55.8	9.2	−28.5	28.7	57.4	275.0	13.2	9.4	−57.4	95.0
						LEF		Decl °	Incl °	α95 °	k	
							Sect. mean	320.8	−62.6	10.0	85.1	
								Lat °N	Long °E	A95 °	K	
							Mean pole	−55.70	081.8	14.4	41.80	N = 4

Upper Elliot Formation (UEF)					Section location		Pole location				VGP	
Section #	n	Decl °	Incl °	α95 °	Lat °N	Long °E	Pole lat °N	Pole long °E	dm	dp	Lat °N	Long °E
BP UEF	46	298.8	72.6	5.8	31.4	27.3	41.3	245.6	10.3	9.1	41.3	65.6
QUT UEF	28	341.0	58.4	5.7	30.4	27.6	72.2	263.2	8.4	6.3	72.2	83.2
MAS UEF	20	328.0	46.4	8.7	29.3	27.5	61.9	292.9	11.2	7.2	61.9	112.9
DAM UEF	38	316.7	60.2	6.1	29.2	27.3	53.1	266.6	9.2	7.0	53.1	86.6
GGP UEF	16	319.4	56.5	8.3	28.5	28.7	55.1	273.8	12.0	8.7	55.1	93.8
GGB UEF	14	345.9	55.8	15.5	28.5	28.7	75.8	261.7	22.2	15.9	75.8	81.7
LIK UEF	18	4.4	63.6	8.5	29.3	27.5	73.7	196.4	13.5	10.7	73.7	16.4
LIKE UEF	12	300.7	61.7	9.4	29.9	27.3	41.6	264.7	14.5	11.2	41.6	84.7
OLD UEF	12	317.7	61.2	12.4	29.1	27.4	53.6	264.5	19.0	14.6	53.6	84.5
SEE UEF	23	143.1	46.9	8.4	29.8	27.5	57.8	111.2	10.8	7.0	57.8	111.2
						UEF		Decl °	Incl °	α95 °	k	
							Sect. mean	327.7	62.7	26.9	4.2	
								Lat °N	Long °E	A95 °	K	
							Mean pole	−60.3	85.2	10.1	23.7	N = 10

The mean palaeomagnetic pole position (Fig. 3 and Table 2) is indicated at latitude 55.7°S during the deposition of the LEF in comparison to the 60.3°S during the UEF's deposition. The overall Elliot Formation (EF) palaeopole position is indicated at latitude 60.5°S. The palaeopole calculated here for the EF is comparable to that of McElhinny and McFadden (2000) for the Early Jurassic of Africa, i.e. EF (this study): 60.5°S, 082.6°E vs. African Early Jurassic: 68°S, 247°E and De Kock (2003) Elliot Formation pole of 68.9°S, 265.6°E. The Late Triassic African palaeopole sits at 65.8°S and 259.7°E (McElhinny and McFadden, 2000).

Fig. 3 illustrates the geographical setting of the mean palaeomagnetic pole calculated in this study for the combined Elliot Formation (EF), LEF and UEF. Comparable, previously published palaeopoles for South America and Africa, in addition to the Gondwana apparent polar wander path, are listed in Table 2 and illustrated in Fig. 4.

For comparison with global palaeopoles of similar time periods, West Africa was used as the reference plate to which other continental plates (in this case, South America) were restored. Southern African co-ordinates were rotated to West African co-ordinates by −7.8° around an Euler pole situated at 9.3°N and 5.7°E. South American co-ordinates were rotated to West African co-ordinates about an Euler pole situated at 53°N and 325°E. Fig. 4 illustrates the apparent polar wander path (APWP) with reference to the placement of the Elliot Formation as per this study and that of De Kock (2003). Fig. 4 also depicts LEF pole positions with those of a similar assumed age (Late Triassic) from Africa and South America in addition to the true polar wander path for ~215 Ma of Iglesia llanos and Prezzi (2013). The mean LEF palaeopole and confidence interval overlaps most significantly with the Los Colorados Formation at ~210 Ma and the Foum-Zguid dykes of Morocco (200 Ma) as well as the African Late Triassic sedimentary rocks (~221 Ma) and Suriname dolerite dykes (~227–237 Ma; Fig. 4). Two Los Colorados Formation palaeopoles are plotted in Fig. 4. Palaeopole #2 is by Iglesia llanos and Prezzi (2013) and #4 is a more recent study by Kent et al. (2014a). The positioning of the original ~210 Ma pole (#2; Fig. 4) shows considerable overlap with the LEF palaeopoles of BP and DAM.

3.2. Stratigraphic correlation

In this study, magnetozones are defined as two or more successive samples of the same polarity. They were designated EF (Elliot Formation) and grouped and numbered. Unconstrained intervals (marked

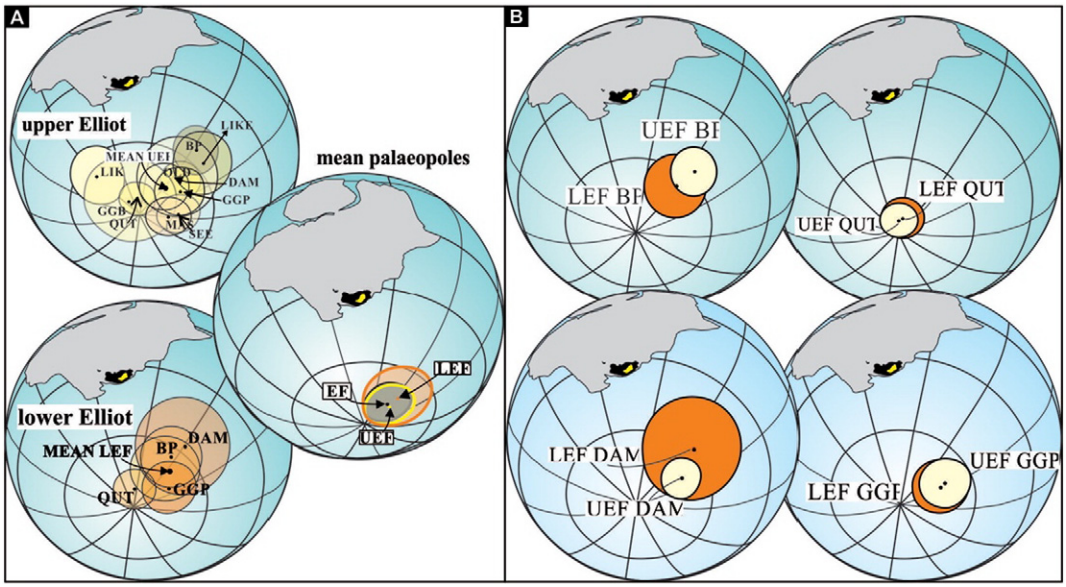


Fig. 3. (A) Mean LEF, UEF and Elliot Formation (EF) palaeopoles. Section abbreviations: BP: Barkly Pass, QUT: Quthing, LIK & LIKE: Likhoele Mountain; SEE: Seeiso; MAS: Maseru; DAM: Damplaats, OLD: Oldenburg, GGP &GGB: Golden Gate Highlands National Park; (B) LEF (orange) and UEF (cream) confidence intervals for Barkly Pass (BP), Quthing (QUT), Damplaats (DAM) and Golden Gate National Park (GGP). The outline of the main Karoo Basin within Gondwana is in black and Stormberg Group is in yellow. Southern African co-ordinates rotated to West African co-ordinates about an Euler pole that is situated at 9.3°N and 5.7°E and rotates through an angle of −7.8° to restore East Gondwana.

grey) are the result of either no sample collected or samples with secondary present field overprint. The composite magnetostratigraphic section of the Elliot Formation was constructed and scaled by correlating normal and reverse magnetozones defined by all study sections (Fig. 5). The resulting composite Elliot Formation section comprises 7 polarity pairs (EF2–8), and two single polarity intervals (the LEF EF1r and the UEF EF9n), encompassing a ~280 m thick composite section.

Composite magnetozones may be thinner or thicker at individual sections given isopach changes due to local erosional events and variation in depositional conditions, which are common within continental, especially fluvio-lacustrine, environments (see Materials and methods). Thus, the polarity pairs in the composite section are idealised and the composite section should be used as a guide for comparison and framework for future magnetostratigraphic work in the Elliot Formation. The Elliot Formation paraconformity signified by the regional lithostratigraphic transition from the LEF-UEF (Bordy et al., 2004b) represents a stratigraphic gap of unknown duration. Estimation of the missing time will have bearing on the correlations provided here. No radiometric dates have been used to tie the section, although the Ar-Ar age of ~215 Ma Cape Fold Belt orogenic pulses (Hälbich, 1983) has been considered as a guide for the unconformable contact of the Molteno-Elliot Formations (Bordy et al., 2005).

The Barkly Pass (BP), Quthing (QUT) and Golden Gate National Park (GGP) study sections represent the longest sampled LEF sections and aided in detailing the LEF magnetic polarity zonation patterns across the basin. The thickest exposures of the LEF occur in the southern part of the basin and therefore magnetic polarity patterns (as captured by sections BP and QUT) are more detailed than in comparison to the more condensed northern sections. One, up to ~75 m, long reverse magnetic chron (EF1r) and four polarity pairs (EF2–5) were identified in the LEF (Fig. 5). This compliments and supplements the two chrons (EF1r–EF2n) described by de Kock (2003).

The Barkly Pass LEF sampled section, as a composite with the original sampling of De Kock (2003), represents an almost complete magnetic polarity zonation of the holostratotype (Figs. 5, 6). The remaining central and northern sections at Likhoele (LIK), Likhoele east (LIKE), Maseru (MAS), Damplaats (DAM) and northernmost section Golden Gate National Park (GGP) all record a lowermost normal LEF polarity zone, often overlain by poorly constrained intervals which transition across the uppermost LEF and lithostratigraphic LEF-UEF contact. The

Table 2
Selected Upper Triassic-Lower Jurassic palaeomagnetic poles from Africa and South America. Lat S, Long E, A95 = latitude, longitude and semi angle of 95% confidence (adapted from Iglesia Llanos and Prezzi, 2013).

	Mean age (Ma)	Lat. S	Long. E	A95	Reference	Ref#
Africa						
Upper Triassic Sed.	221.5	54.9	43.3	11.5?	Ghorabi and Henry, 1991	3020
Foum-Zguid Formation	200	67.9	67.9	3.9?	Palencia-Ortas et al., 2011	–
Zarzaitine Fm.	206.5	70.9	55.1	2.6?	Kies et al., 1995	2932
Karroo Combined	198	65.4	75.1	9.5?	McElhinny and Jones, 1965	317
Karroo Igneous complex	186	62.9	98.3	3.0?	Hargraves et al., 1997	3430
Marangudzi Complex.a	186	70.7	106.7	8.7?	Brock, 1968	470
Drakensberg Group	180–184	75.2	276.4	5.8	Moulin et al., 2011	–
'Stormberg Lavas'	185	71.6	93.5	3.2?	Kosterov and Perrin, 1996	3090
South America						
Los Colorados Fm.	210	76	280	8.0?	Vizañ et al., 2004	–
Anari-Tapirapua Fm.	196.6 ± 4	65.5	250	3.5?	Montes-Lauar et al., 1994	3316
Neuquén Basin	197	51	223	6.0?	Iglesia Llanos et al., 2006	–
Lepa-Osta Arena Fm.	180–186	75.5	129.5	6.0?	Vizañ, 1998	3314
Neuquén Basin	185	74	67	5.0?	Iglesia Llanos et al., 2006	–

lowermost extent of the LEF was very difficult, to impossible, to sample at these sections, and therefore comparison to the lowermost LEF parts from the BP, QUT and GGP sections is not possible.

The UEF is characterised by a pattern of 3 normal-reverse polarity intervals, capped by a single normal polarity zone, labelled EF6, EF7, EF8 and EF9n respectively (Fig. 5). The similarity in EF6-EF7 polarity pattern for the UEF across the basin illustrates the palaeomagnetic stability of the Elliot Formation, as noted by De Kock (2003).

4. Discussion

4.1. Pole comparisons

Mean LEF, UEF and combined Elliot Formation (EF) poles show overlap with the Triassic-Jurassic True Polar Wander (TPW; Iglesia Llanos and Prezzi, 2013). The LEF palaeopole overlaps with the APWP for the Late Triassic and Early Jurassic as well as several Late Triassic Argentinian poles (e.g., Los Colorados Formation; Iglesia Llanos and Prezzi, 2013). Importantly, the earliest radiometrically dated CAMP volcanic rocks (i.e., the North Mountain basalts, ~201.6 Ma; Blackburn et al., 2013) have a palaeopole at −71.9°S and 68.2°E ($\alpha 95 = 10.7$; Torsvik et al., 2012) which is similar to LEF pole position.

The combined Elliot Formation and UEF poles are consistent with previous Early Jurassic palaeopoles (Fig. 4) and these poles show relatively good correspondence with Formations of suspected similar age, i.e., the northern Patagonian Anari-Tapirapua Formation in the Neuquén Basin (~197 Ma) as reported by Iglesia llanos and Prezzi (2013). Palaeopoles of the Lower Jurassic North American Moenave Formation (~197 Ma; Donohoo-Hurley et al., 2010) are rotated to southern Africa in Torsvik et al. (2012) and plot at latitude −73.2°S and longitude 55.4°E ($\alpha 95 = 2.8$). The geographic location of this palaeopole is comparable to that reported for the Hartford Newark Basin volcanics (−65.4°S, 69.2°E, $\alpha 95 = 4$; Torsvik et al., 2012), and is more closely comparable to the mean EF and UEF palaeopole (Table 1).

Little overlap occurs between the Elliot Formation poles and those calculated for the Drakensberg Group basalts by Moulin et al. (2011). However, overlap does occur with poles calculated for the 'Karoo Igneous' of Iglesia llanos and Prezzi (2013) and the previous Elliot Formation magnetostratigraphic study by De Kock (2003). The overlap between the 95% confidence circles of the current study UEF and combined EF with the De Kock (2003) Elliot Formation pole is explained by the UEF sampling bias of De Kock (2003). This, moreover, may also argue that the secondary late Early Jurassic (~Toarcian) overprint is indistinguishable from the primary remanent magnetisation.

4.2. Global correlations

The southern African Upper Triassic-Lower Jurassic Elliot Formation was magnetostratigraphically compared to the continental sections of the North American Newark APTS, the Hartford Basin, the Chinle and Moenave Formations of the Colorado Plateau (e.g., Nevada, Utah, northern Arizona, western New Mexico), the Argentinian Ischigualasto–Villa Union Basin as well as the European St. Audrie's Bay (UK) and Paris Basin (France) (see Fig. 7 for ages and references).

The relative age assignments of the Elliot Formation are based on its stratigraphic position in the Stormberg Group and its diverse vertebrate fossil assemblage. The latter being suitable for faunal comparisons with other global Upper Triassic-Lower Jurassic sections, although most sections are lacking a robust chronostratigraphic framework. In the case of the LEF, the taxa are generally considered to be Norian (Lucas and Hancox, 2001; Knoll, 2004), and thus are contemporaneous with those of the middle Norian in Germany, the Los Colorados Formation (Ischigualasto–Villa Union Basin) in Argentina and the uppermost Chinle Formation in the USA (Yates and Kitching, 2003; Yates, 2007a). Recent radiometric dates from these formations and their magnetostratigraphic tie to Newark-APTS 2010 were useful in the global assessment of the LEF magnetozones (Fig. 7).

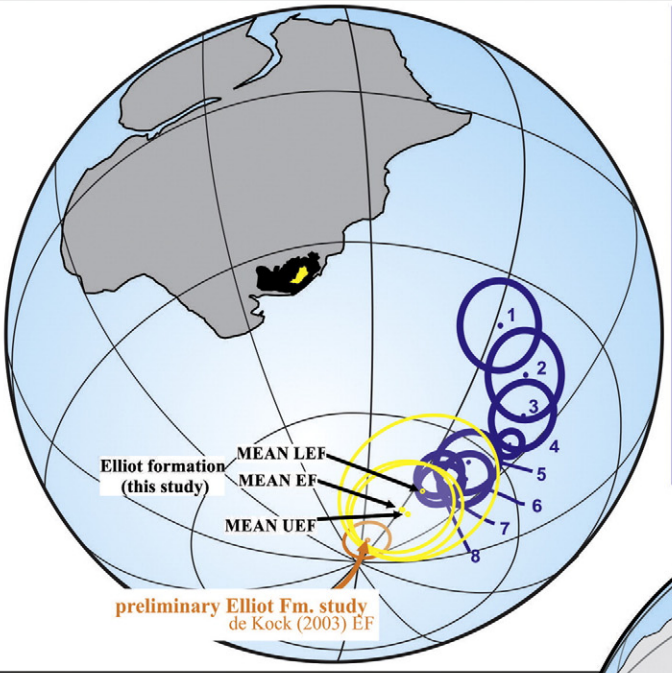
Typically, the LEF has been correlated with the Los Colorados Formation on the basis of their comparable sauropodomorph fauna, which also shares several affinities with Europe (Yates, 2007b; McPhee, 2013). The uppermost (upper third; Kent et al., 2014a) of the Los Colorados Formation is typified by the La Esquina local fauna, which contains the sauropodomorph dinosaurs *Riojasaurus* and *Coloradisaurus* (Arcucci et al., 2004). The LEF sauropodomorph *Eucnemesaurus fortis* is considered by Yates (2007b) as the sister taxon to *Riojasaurus* (Yates, 2007b; McPhee et al., 2015); while the Los Colorados Formation *Lessemsaurus* has been regularly resolved as the sister-taxon to the LEF *Antetonitrus* (Pol and Powell, 2007). The biostratigraphy and biogeography of these formations are still under investigation, and although phylogenetic relatedness of various taxa does not imply equivalence in age for the formations, it does provide a basis for a more meaningful comparison.

Currently, age estimates of the Los Colorados Formation range from ~227 to ~213 Ma based on magnetostratigraphic correlation with the Newark magnetochrons E7r and E15n (Kent et al., 2014a). The sole radiometric controls for the Los Colorados Formation come from $^{40}Ar/^{39}Ar$ dating of the underlying beds of the upper Valle de la Luna Member (uppermost Ischigualasto Formation; Fig. 8) and these provide a discrepancy in ages (Fig. 8) despite being obtained from the same unit. The age discrepancy is likely to indicate either the low-precision methods of $^{40}Ar/^{39}Ar$ dating, the complexity in underpinning a central age-date plateau, or other sedimentological complications such as a strongly diachronous Formation with many paraconformities which are difficult to detect in isolated outcrops.

The age range of the Los Colorados Formation is complimented by the mix of Late Triassic and Early Jurassic vertebrate forms (see Arcucci et al., 2004) which implies several undetected paraconformities in the Los Colorados Formation (possibly between the La Sal and La Esquina sections). A younger age for the Los Colorados Formation has been suggested previously (e.g., Lucas, 1998; Arcucci et al., 2004) on the basis of its fauna and without absolute age controls. For these reasons, an alternative correlation of the Los Colorados Formation is considered here, which would shift the unit between ~218 Ma (E13.1r) and ~208 Ma (E18n; Fig. 8B). In this attempted correlation (Figs. 7, 8A, B), the Los Colorados Formation overlaps with the Sonsela Member of the Chinle Formation (USA).

A more conservative correlation of the lowermost LEF long reversal (EF1r) with Los Colorados Formation magnetozone LC7, and correspondingly with the Newark subchron E14r (~214–213 Ma) as per Kent et al. (2014a), is illustrated in Figs. 7 and 8A. This can be tentatively linked to the uppermost reversal within the Sonsela Member of the Chinle Formation. The Black Forest Bed (Petrified Forest Member, Chinle Formation) has been dated to 209.93 Ma (Ramezani et al., 2011) (Figs. 7, 8A), and potentially places the lowermost LEF into the Norian at ~213 Ma. This age is in turn in surprisingly good agreement with the age of the unconformable contact between the Molteno and Elliot Formations, which is assumed to be ~215 Ma (Catuneanu et al., 1998; Bordy et al., 2004b, 2005). In summation, hypothesising a younger Triassic age for the LEF provides a robust tie with the current understating of the geology of the main Karoo Basin and the evolutionary tempo of its sauropodomorphs.

Given the above arguments, it is likely that the EF1r magnetozone in the lowermost LEF ties with either the LC5r or the LC7r zones in the uppermost Los Colorados Formation regardless of its correlation to the Newark APTS (Fig. 8B). This is due to the palaeomagnetic ties and the comparison of the stratigraphic occurrence and assumed phylogenetic closeness of the prosauropod dinosaurs *Riojasaurus* and LEF *Eucnemesaurus*. The stratigraphic position of *Eucnemesaurus* gives credence to a younger age for the LEF (<~215 Ma; Fig. 8B). It is difficult to narrow down which of these correlations is more accurate due to the lack of radiometric dates. Considering the Kent et al. (2014a) model (Fig.

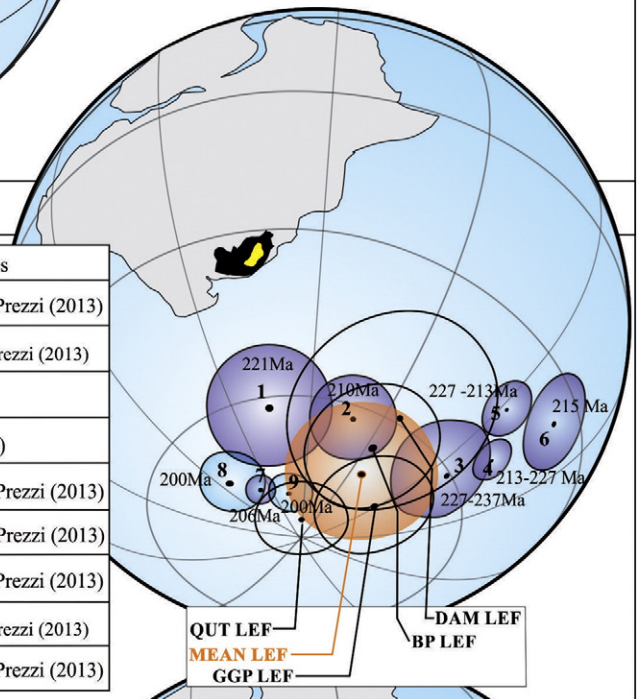


Apparent Polar Wander Path (APWP)

1 Gondwana Early Permian
2 Gondwana Mid Permian
3 Gondwana L. Permian - E. Triassic
4 Gondwana L. Jurassic - E. Cretaceous
5 Gondwana L. Jurassic
6 Gondwana L. Triassic
7 Gondwana E. Jurassic
8 Gondwana M. Jurassic

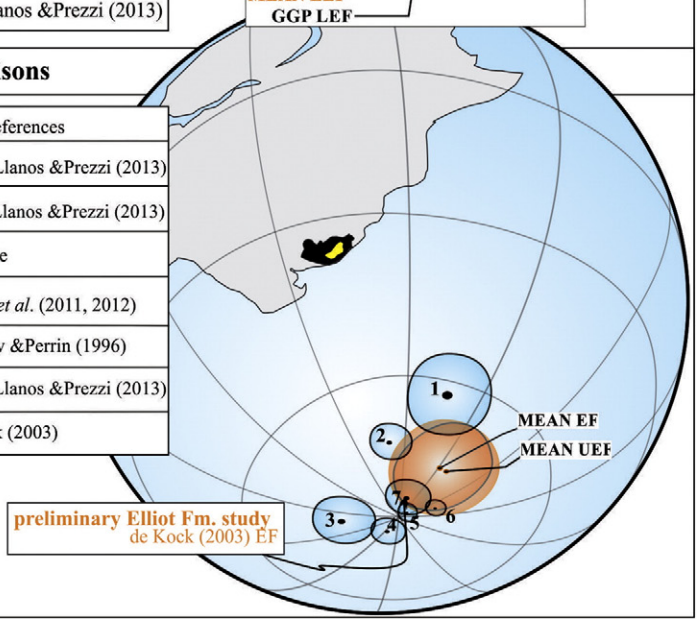
lower Elliot Formation (LEF) comparisons

No.	Unit	Age (Ma)	References
1	African Upper Triassic Sediments	221	Iglesia Llanos &Prezzi (2013)
2	Los Colorados Fm. (Argentina)	210	Iglesia Llanos &Prezzi (2013)
3	Dolerite Dykes (Suriname)	237-227	Database
4	Los Colorados Fm.	213-227	Kent *et al.* (2014)
5	Triassic - Jurassic TPW	227-213	Iglesia Llanos &Prezzi (2013)
6	Triassic - Jurassic TPW	215	Iglesia Llanos &Prezzi (2013)
7	Zarzaitine Fm. (SE Algeria)	206	Iglesia Llanos &Prezzi (2013)
8	CAMP lavas (Morocco)	200	Iglesia Llanos &Prezzi (2013)
9	Foum-Zguid dyke (Morocco)	200	Iglesia Llanos &Prezzi (2013)



upper Elliot Formation (UEF) comparisons

No.	Unit	Age (Ma)	References
1	Los Colorados Fm. (Argentina)	210	Iglesia Llanos &Prezzi (2013)
2	Neuquén Basin	~197	Iglesia Llanos &Prezzi (2013)
3	Anari-Tapirapua Fm.	196.6±0.4	Database
4	Drakenberg Group	~179.2 ± 1.8 - 184.8 ± 2.6	Moulin *et al.* (2011, 2012)
5	Stormberg lavas/Drakenberg Grp.		Kosterov &Perrin (1996)
6	Karoo Igneous	~186 Ma	Iglesia Llanos &Prezzi (2013)
7	Preliminary study: Elliot Fm.	Early Jurassic	de Kock (2003)



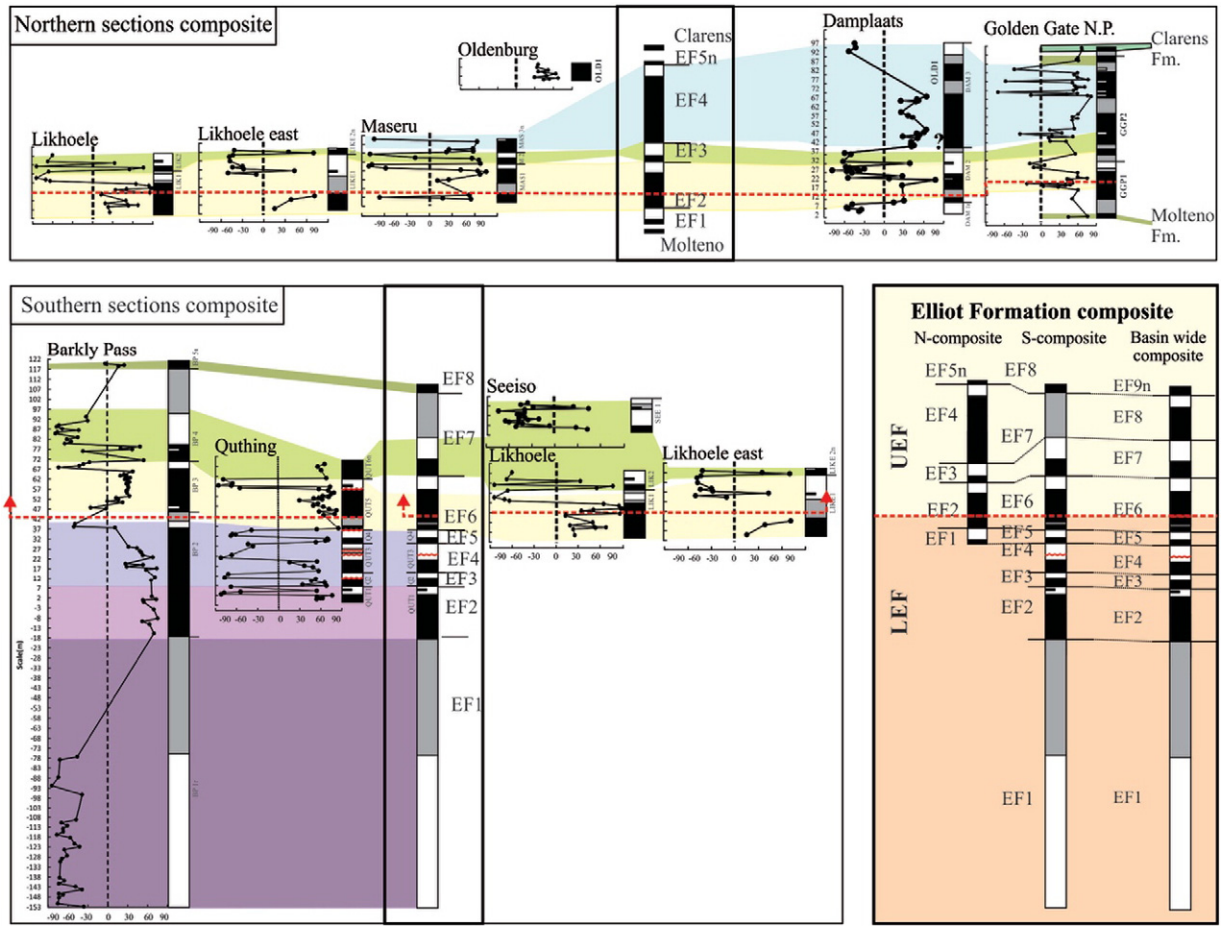


Fig. 5. Magnetostratigraphic sections and their relative composites from the southern and northern part of the main Karoo Basin as well the composite magnetostratigraphic section of the Elliot Formation. The red dashed line represents the boundary between the lower and upper Elliot Formations (lithostratigraphic LEF-UEF contact).

8A) as the most correct correlation, if EF1r ties more successfully based on fossil occurrences with LC5 (=E11/12) it would set the base of the Elliot Formation to ~218/219 Ma. If EF1r is tied to LC7r (= E14r) this would place the base of the Elliot Formation at ~215 Ma (Fig. 8A). Given previous, if contentious, age estimates for the onset of LEF sedimentation at ~215 Ma (Catuneanu et al., 1998; Bordy et al., 2004b, 2005), the ~214 Ma age tie (given E14r = LC5 = EF1r; Fig. 8B) for the lowermost LEF is, however, also probable based on fossil occurrences. Moreover, the high resolution chronostratigraphic framework of the continental Chinle Formation closely and independently ties it to global Upper Triassic marine beds and places the appearance of dinosaurs in North America at ~223 Ma (Irmis et al., 2011; Ramezani et al., 2011; Atchley et al., 2013; Ramezani et al., 2014). This is considered here to further support a slightly younger Triassic age for the sauropodomorphs in the LEF.

Other possible correlations between the formations could pivot on theropods, such as *Coelophysis bauri*, which in the Chinle Formation co-occurs with basal saurischians and rauisuchians (Parker and Martz, 2010). This is similar to the upper parts of the Los Colorados Formation, where there are pseudosuchians such as *Riojasuchus tenuisceps* (von Baczko and Desojo, 2016), and where the basal theropod is a coelophysoid equivalent, *Zupayasaurus rougieri* (Ezcurra and Novas, 2007). In South Africa and Lesotho, there are limited *Coelophysis*-like remains and mostly constrained to the UEF as well as the overlying Clarens Formation (Kitching and Raath, 1984). However, *Grallator* trackways referable to (coelophysid) theropods are found from the LEF and into the UEF and Clarens Formation (Olsen and Galton, 1984; Knoll, 2005; Sciscio et al., 2016).

The position of LEF magnetochrons EF2 and EF3 may be placed at two intervals of disparate age within the Newark APTS. Firstly, biostratigraphically the EF2 and EF3 magnetochrons can be tied to the upper third of the Los Colorados Formation (LC8n and the Newark magnetochron E15, ~213 Ma) despite the gap in sampling between EF1r and EF2 (Fig. 7). A rapid rate of sedimentation over the unknown LEF interval would be consistent with this interpretation, but which cannot be sedimentologically supported at Barkly Pass. Based on an assessment of Kent et al. (2014a) model (Fig. 7), the EF2 and EF3 could, based on the correlation with EF1r to LC7r, tie into the Newark E15/16 and E17, respectively. Relative to the Chinle Formation, this would place EF2 and EF3 into the Black Forest Bed and Petrified Forest Member, respectively (~209–~207 Ma; Fig. 7). This is solely based on polarity for the correlation to the Newark basin APTS.

In considering a younger age for the Los Colorados Formation as presented in Fig. 8B, the EF2 and EF3 magnetochrons could tie, based on

Fig. 4. Palaeomagnetic poles with 95% confidence circles for mean Elliot Formation (EF; orange), LEF and UEF (this study), Elliot Formation (De Kock, 2003) and previously published Apparent Polar Wander Paths (APWP), and Late Triassic-Early Jurassic palaeopoles with their confidence limits (purple = Late Triassic; blue = Early Jurassic). Toarcian Drakensberg Group basalts are indicated. The outline of the main Karoo Basin within Gondwana is in black and Stormberg Group is in yellow. Southern African co-ordinates are rotated to West African co-ordinates about an Euler pole that is situated at 9.3°N and 5.7°E. South American co-ordinates rotated to West African co-ordinates about an Euler pole that is situated at 53°N and 325°E.

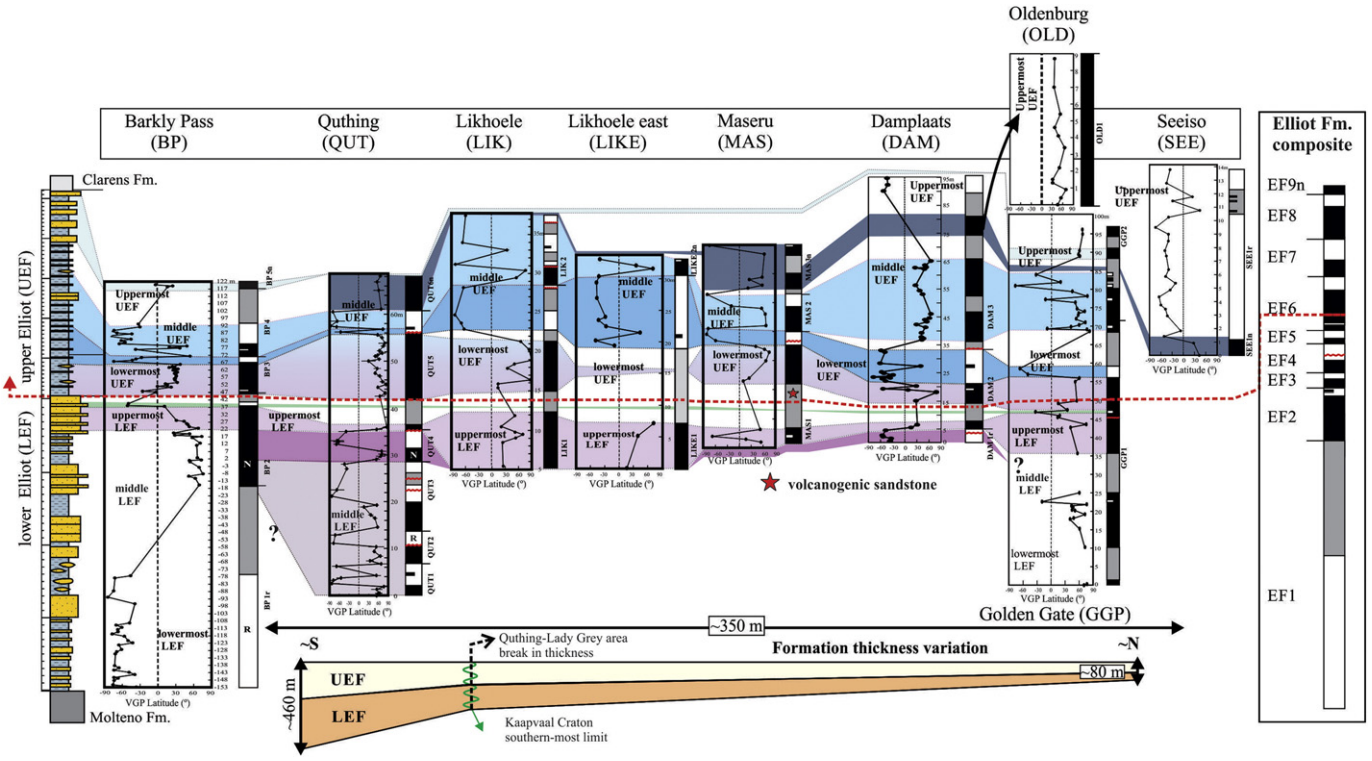


Fig. 6. Correlations and composite magnetic stratigraphy of the Elliot Formation from sections at Barkly Pass, Quthing, Likhoele and Likhoele east, Seeiso, Damplaats, Oldenburg and Golden Gate National Park (GGP and GGB). The red dashed line represents the boundary between the lower and upper Elliot Formations (lithostratigraphic LEF-UEF contact). Wavy red lines indicated on individual polarity logs are erosional surfaces identified in the field. Thickness variation of the Elliot Formation within the main Karoo Basin is indicated at the base of the figure with the schematic S-N cross-section. Idealised sedimentological log of the Elliot Formation is adapted from Bordy and Eriksson (2015).

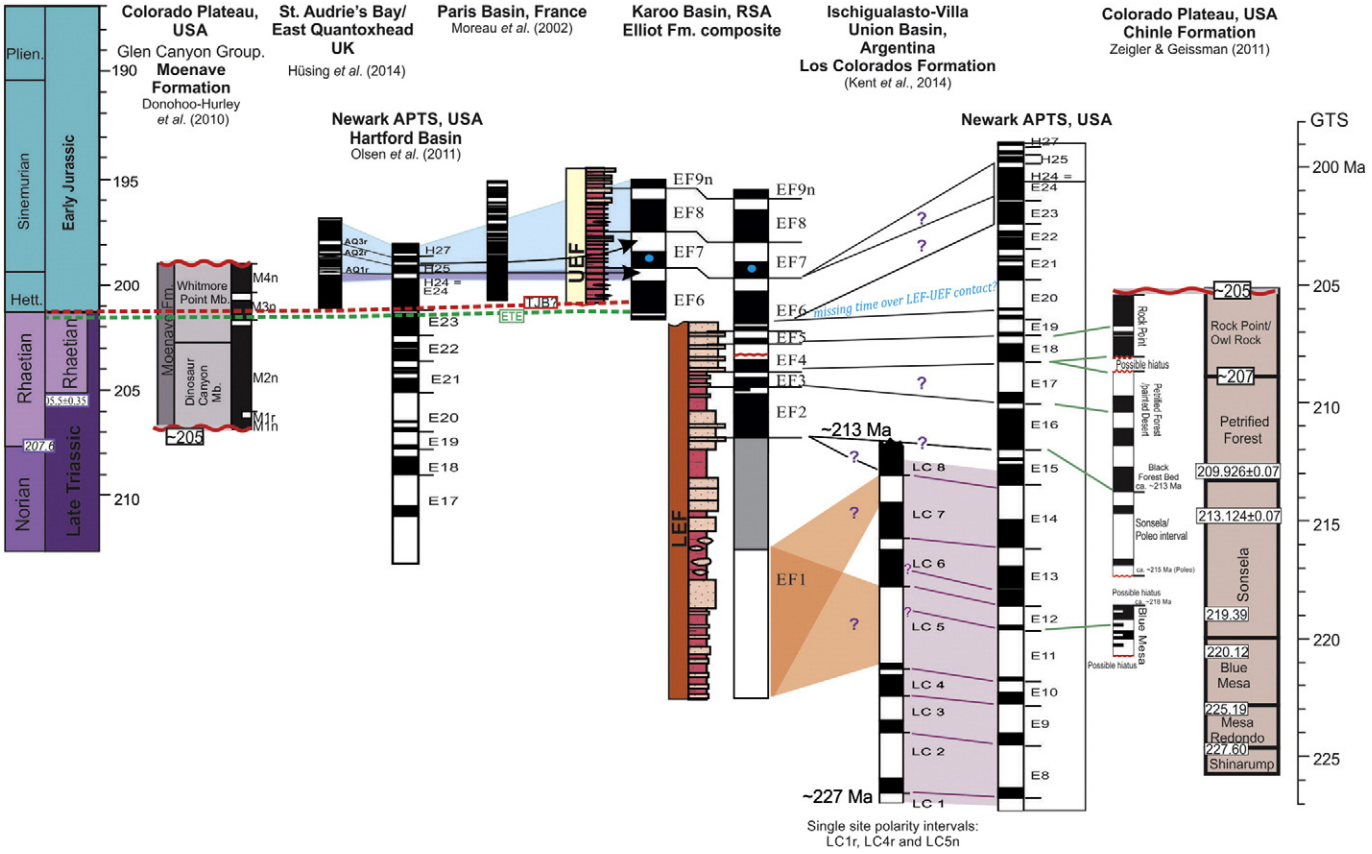


Fig. 7. Magnetostratigraphic correlation of the Upper Triassic-Lower Jurassic Elliot Formation to continental and marine magnetostratigraphic sections from North America (Newark APTS, Hartford Basin, Colorado Plateau), South America (Argentinian Ischigualasto–Villa Union Basin) and Europe (St. Audrie's Bay, UK; Paris Basin, France).

magnetostratigraphy alone and an estimate rate of ~2 reversals per million years, into EF2 = LC8n = E18/E19 and EF3 into E20. This would loosely tie to the Rock Point/Owl Rock Member (Zeigler and Geissman, 2011) with the latter considered to span from ~207.2 Ma (Petrified Forest and Owl Rock contact) to 205 Ma (Ramezani et al., 2011, 2014). Thus, the EF2n may represent a lowermost age of ~207 Ma and the EF3r may represent an uppermost age of ~205 Ma. This would make the stratigraphic gap between the EF1r and EF2n ~2 to 6 million years. These units have not been shown to be correlatives of those found in the main Karoo Basin, and chronologically and biostratigraphically it is difficult to link the Rock Point/Owl Rock Member with the LEF. Conchostracans are the only fauna which have been reported in the LEF (e.g., *Estheria* sp., at Mamokoaqo; Ellenberger, 1970) and are also present in the Late Norian Rock Point Member (Lucas et al., 2011). Biostratigraphic support for the correlation with the Rock Point/Owl Rock Member of the Chinle Formation is weak as the Revueltian-aged (early to mid-Norian) taxa (e.g., *Pseudopalatus*, *Typothorax*, *Trilophosaurus* and metoposaurs) from the Owl Rock Member have not proven fruitful for correlation (Parker and Martz, 2010); and furthermore, they have no sister taxon in the Elliot Formation.

The marine section at Pignola-Abriola places the Norian-Rhaetian boundary (MPA 5r magnetozone) at ca. 205.7 Ma (Maron et al., 2015) within the Newark APTS magnetozone E20r.2r directly below the proposed correlation here (of EF3) to Newark magnetochron E20.

Conversely, the current models cited in Fig. 8 would allow for the aforementioned correlation of EF1r to the Los Colorados Formation LC5 and thus polarity interval EF2 can, theoretically, fall between magnetochrons LC6–LC8n because of bio- and chronostratigraphic constraints. That said, the sampling gap between EF1r and EF2 must represent a significant amount of time given the width of the unknown polarity interval (~60 m). EF2 represents the lowermost middle section of the LEF and the last occurrences of *Melanorosaurus*, *Eucnemesaurus* and the capitosaurids, to date (Fig. 9). It could convincingly tie with the LC8n of the Los Colorados Formation and be tentatively linked to the Newark E18 and E19 magnetochrons based on these specimens.

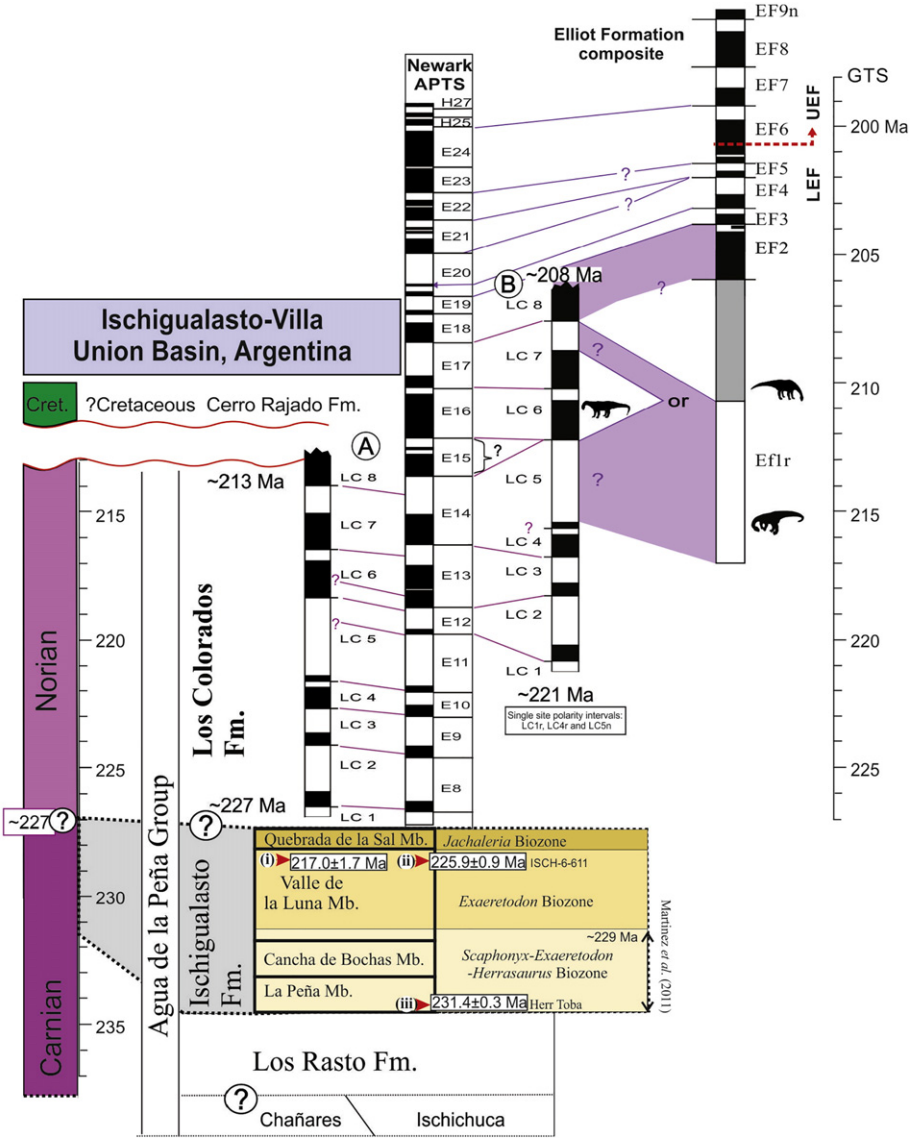


Fig. 8. Potential magnetostratigraphic correlations of the Argentinian Los Colorados Formation (Agua la Pena Group) to the Newark APTS2010 based on: A) Kent et al. (2014a) which suggest that the base of the Formation is at ~227 Ma; and B) new palaeontological and magnetostratigraphic work in the LEF. The underlying Ischigualasto Formation chronostratigraphy is based on: (i) $^{40}Ar/^{39}Ar$ date of the Valle de la Luna Member (feldspars): 217.0 ± 1.7 Ma by Shipman (2004); (ii) $^{40}Ar/^{39}Ar$ date of the Valle de la Luna Member (ISCH-6-611): 225.9 ± 0.9 Ma by Martínez et al. (2011); (iii) $^{40}Ar/^{39}Ar$ date of the La Peña Member (Herr Toba bentonite): 231.4 ± 0.3 Ma (1σ total error) by Martínez et al. (2011).

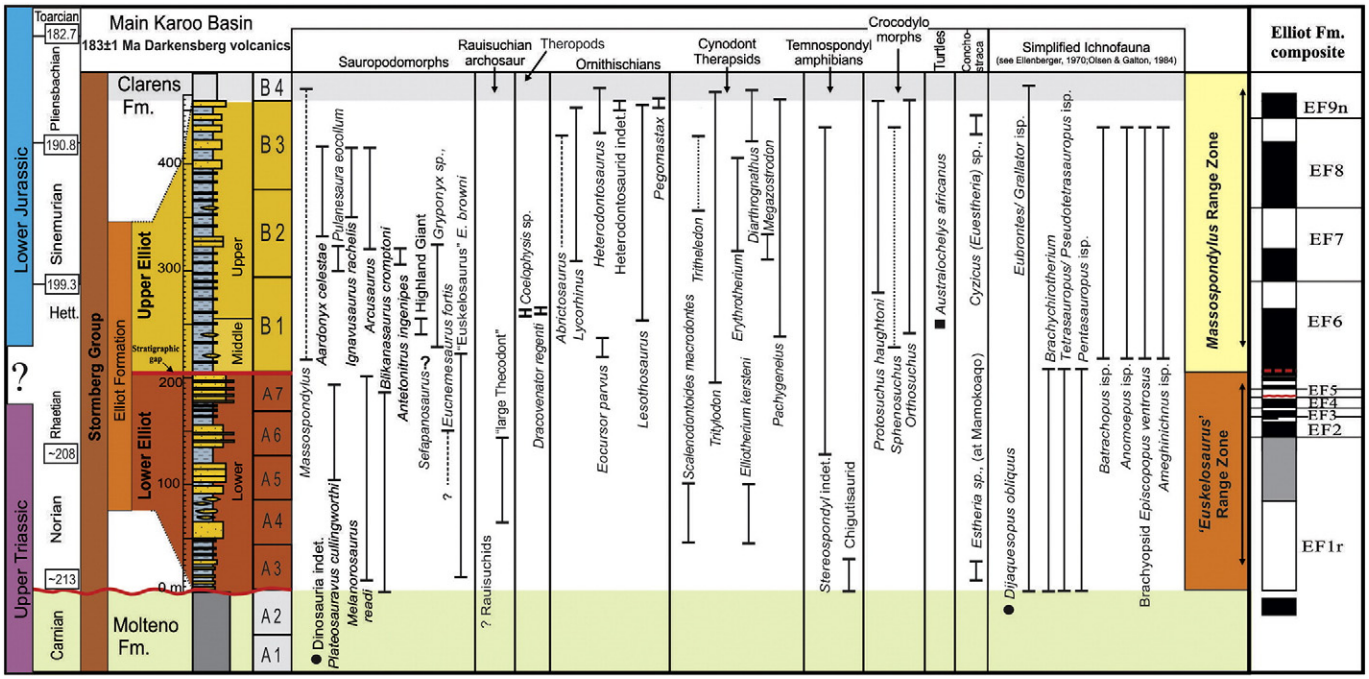


Fig. 9. Biostratigraphy of the Elliot Formation based on revisions of Kitching and Raath (1984), Olsen and Galton (1984), Lucas and Hancox (2001), Bordy et al. (2004a) and Knoll (2004, 2005) in relation to the current composite magnetostratigraphic section of the Elliot Formation. The simplified sedimentary log is based on the holostratotype section of the Elliot Formation in the Barkly Pass (Eastern Cape; S31°15′23″, E27°49′45″; modified from Bordy and Eriksson, 2015). Biostratigraphic range zones of Kitching and Raath (1984) and Zones A1–B4 of Ellenberger (1970) are shown for reference.

This would delimit a time span between 207 and 206 Ma for EF2. It would, additionally, cautiously place EF3 into the overlying Newark chron E20, and likely denote the E20.1n and E20.1r (Fig. 8B).

Fig. 8B presents correlations for EF4, EF5 and EF6 largely based on the younger age of the Los Colorados Formation and its correlation with EF1r–EF3, thus linking the latter magnetozones into the Newark E20–E23. Once more, uncertainty around these correlations stems from the lack of comparable fossil material or radiometric dates. The correlation of EF4 to E20.2n and E20.2r is done on the basis of combining EF3 and EF4 into one chron, which can be reasonably attempted based on the homogeneity in Elliot Formation fauna across these magnetozones (Fig. 9). EF5 can be tentatively placed within the E21–E22, and if Elliot Formation magnetozone EF5r corresponds with Newark E22r then the correlation of Donohoo-Hurley et al. (2010) for the Late Triassic Moenave Formation M1r (on the base of polarity pattern) can be utilised. This Moenave Formation magnetozone (M1r), as well as the overlying M2r (correlated to Newark E23r), have been tied with changes in the palynoflora (specifically *Classopollis meyerianus* to *Classopollis torusus*) and the ichnogenus *Eubrontes* to the Newark APTS. Moenave Formation M2r has been considered to be correlated with Newark chron E23r, SA5n.2r or SA5n.3r of St Audrie's Bay (UK), the H- from the section at Oyuklu (Turkey) as well as BIT5n.1r in the Italcementi Quarry (Italy; Donohoo-Hurley et al., 2010). Lucas et al. (2011) have proposed that the M1r is Late Norian in age based on the occurrence of the Rhaetian monospecific *Euestheria brodieana* conchostraca assemblage above the M2r interval and thus, that the M2r interval represents the TJB. This is based on the lithostratigraphic correlation of conchostracans from the Whitmore Point Member (Kozur and Weems, 2010; Lucas et al., 2011). In contrast, Kirkland et al. (2014) have provided evidence for the ETE to occur over the M1 and M2 intervals with the TJB occurring over the M3 and M4 polarity (Fig. 7). This would tie in well with the magneto- and biostratigraphy of the Elliot Formation whereby the ETE would be more likely to coincide with the LEF-UEF contact (Figs. 7, 8). From a southern African perspective the fauna which may aid comparisons, specifically the genus *Euestheria* (*Cyzicus draperi*; Ellenberger, 1970; Tasch, 1984) and the ichnogenus *Eubrontes* (loosely synonymised with *Grallator*; Olsen and Galton, 1984, Lucas and Hancox, 2001), are both present in the LEF and UEF (Fig. 9) and therefore do not aid in refining correlations or biostratigraphic ages.

The normal magnetozone EF6n occurs over the LEF-UEF contact, the latter of which has traditionally and tentatively been considered the TJB in the Elliot Formation (Kitching and Raath, 1984, Smith and Kitching, 1997; Figs. 7, 8), because it coincides with a crudely defined shift in representative vertebrate fauna within the Formation. It also represents a change in lithostratigraphy associated with a paraconformity of unknown duration (Bordy et al., 2004b). This stratigraphic gap could be estimated to be as little as 2 million years based on the reversals rate. The previous palaeomagnetic study of the Elliot Formation (De Kock, 2003) illustrated the presence of one long normal polarity zone (EF2n) corresponding to the uppermost LEF and the transition into the UEF. The long normal polarity approximately corresponds to the EF2–EF6 of the current study (Figs. 7, 8). The EF6.1n (av. ~5–15 m thick) is punctuated by a short reversal (represented by 2 samples at BP and one sample at GGP) just below the LEF-UEF contact. EF6.1r is ~3–5 m thick.

The correlation of the EF6 interval with the Newark APTS (E23 + E24) (Figs. 7, 8B) and the Rhaetian-aged (Kirkland et al., 2014) Moenave Formation M2r (Fig. 7) has been linked to uppermost reversal EF6.2r. This link is lacking substantiating evidence and this correlation is purely based on assumptions about the Late Triassic biostratigraphic age of the Elliot and Moenave Formations. The only evidence which could prove favourable for this correlation is the discovery of a rauisuchian maxilla bearing five teeth (SAM PK K383) that was speculated to have been found in the upper Elliot Formation (Nesbitt et al., 2013). The exact stratigraphic occurrence of this find is, however, unknown (Smith, pers. comm.). This would place the E23r within the EF6.1r, and the TJB could be shifted upwards within EF6.2n, which is a protracted normal polarity period typical of the earliest Jurassic or within the EF7n (a thin normal magnetozone). The normal polarity represented in the EF7n chron is likely condensed, and contains numerous palaeosol horizons, fossil remains of *Protosuchus*, *Pachygenelus*,

Megazostrodon, *Erythrotherium* and *Lycorhinus*, in addition to trackways (e.g., the Moyeni locality) of Early Jurassic age (Figs. 7, 9).

Based on the occurrence of similar body and ichnofossils (Ellenberger, 1970, 1972), for instance, a specimen of *Protosuchus*, conchostracans and the ichnotaxon *Eubrontes*, the Moenave Formation magnetochrons M3 and M4 (Donohoo-Hurley et al., 2010) of the Whitmore Point Member tie to the UEF EF7 (and potentially EF8; Fig. 7) chron. This is considering the *Grallator-Eubrontes* track zone and the *Anomoepus*-like trackways as representative of the Early Jurassic globally (Lockley, 1991; Knoll, 2005). Furthermore, *Protosuchus haughtoni* (Busby and Gow, 1984; Gow, 2000) bears similarities with the Dinosaur Canyon Member crocodylomorph *Protosuchus richardsoni* (Sues and Olsen, 2015). *Protosuchus haughtoni* occurs throughout the UEF and has been reported from the lower parts of the UEF, which would be coincident with the (?latest Triassic) EF6/EF7 magnetochron (Fig. 7). Recently, Kirkland et al. (2014) proposed that, although traditionally considered earliest Hettangian at the Dinosaur Canyon Member, this basal crocodylomorph is in latest Triassic strata in Nova Scotia. Conchostracans in the uppermost UEF have not been considered age-diagnostic and appear to exhibit provincialism (Tasch, 1984), despite that these taxa are used for North American-European correlations (Kozur and Weems, 2010).

In contrast, Fig. 6A utilises the recent work of Hüsing et al. (2014), which has established a reputable magnetostratigraphic composite for the Hettangian at St. Audrie's Bay/East Quantoxhead (UK). This new composite repeats the terrestrial pattern of the Hartford Basin (USA) magnetostratigraphy. The replicated long normal period, a characteristic of the Early Jurassic, is punctuated by three reverse chrons and is seen in both marine and continental sections. If these sections are compared to the UEF, it is likely that EF6 contains both the ETE at the LEF–UEF contact and above this, within EF6.2n or higher, the TJB. This would place the EF7 magnetozone within the earliest Jurassic. Fig. 7 attempts to correlate the Elliot Formation EF6.2r–EF9n to Hartford Basin H24r–H27 and thereby AQ1r–AQ3r (of St. Audrie's Bay/East Quantoxhead composite; AQ). This correlation places the UEF EF6.2r–EF9n as Hettangian–Sinemurian in age.

Magnetochrons EF8 and EF9n can then represent a much later time period (Fig. 7) and can be correlated with the Hettangian Hartford Basin and St. Audrie's Bay/East Quantoxhead composite as well as the Late Sinemurian of the Paris Basin (Hüsing et al., 2014). The older age for these UEF magnetochrons may be supported when considering the occurrence of the tritheledontid *Pachygenelus monus* (Gow, 1980) within the stratigraphy. Many *Pachygenelus* specimens have been reported as occurring within ~3 m of the UEF-Clarens Formation contact (Fig. 8; Sidor and Hancox, 2006). The postcanine teeth of *Pachygenelus monus* are similar to *Pachygenelus* cf. *P. monus* from the McCoy Brook Formation (Newark Supergroup), which overlies the U-Pb Zircon SIMS(?) dated North Mountain Basalts (201.566 ± 0.031; Blackburn et al., 2013; Sues and Olsen, 2015). Fedak et al. (2015), however, consider the McCoy Brook Formation to be latest Rhaetian to Early Jurassic in age while it was Tanner and Lucas (2006, 1994) who proposed an Early Jurassic (Hettangian) age. Nevertheless, this could indicate an age of as young as ~201 Ma for the upper part of the UEF.

However, one has to consider the condensed nature of the Elliot Formation in the northern parts of the basin, where *Pachygenelus* specimens have been found, and the potential of this succession to record significant amounts of time. The first assumption would support the extension of the TJB to within the upper UEF and would mean that much of the LEF and lower UEF are Late Triassic in age. The weakness of this reasoning is evident as it is based on one fossil taxon (e.g., *Pachygenelus*), and although an intriguing possibility the occurrences of other contemporaneous fossil groups and their biostratigraphic position, as well as correlation to global localities, is contrary to this idea. Biostratigraphic evidence from South America (Caturrita Formation, Brazil, and Los Colorados Formation, Argentina) of other tritheledontid taxa have been considered Norian in age (Sidor and Hancox, 2006), and thus do not lend support to the above assumption unless ideas of time protracted regional dispersal of fauna are considered.

Morganucodontans are shared between North American, European, Russian, Indian and Chinese basins (Clemens, 1979, 1980; Kielan-Jaworowska et al., 2004). A diversity of Rhaetian-aged morganucodonts have been reported from Saint-Nicolas-de-Port (France; Debuysschere et al., 2015) with *Megazostrodon chenali* sharing similarities with the UEF *M. rudnerae*. Morganucodontans from English Fissure Fills (Mendip Fissures, United Kingdom; Wall and Jenkyns, 2004) extend in age from the Rhaetian to the Early Sinemurian (Whiteside et al., 2016). The UEF morganucodontans *Erythrotherium* and *Megazostrodon*, because of their (comparatively) well-documented locations (Jenkins and Parrington, 1976), could aid the magnetostratigraphic correlation of the UEF with other global sections.

Again, the placement of the EF7–9n magnetochrons into the Early Jurassic seems plausible based on comparative sauropodomorph fossil evidence (Fig. 7). It must be noted that *Massospondylus* may be the eponym for a widespread family of Early Jurassic sauropodomorphs including *Sarahsaurus* (Kayenta Formation; Rowe et al., 2010) and *Leyesaurus marayensis* (Quebrada del Barro Formation; Marayes-El Carrizal Basin, Argentina; Apaldetti et al., 2011). Apaldetti et al. (2011) consider that *Sarahsaurus* is related to the UEF *Ignavusaurus* (Knoll, 2010) and that the Argentinian *Leyesaurus* and *Adeopapposaurus* constitute the sister group to *Massospondylus*. Based on biostratigraphic correlations alone, it would place the *Massospondylus* Range Zone (UEF and lowermost Clarens Formation) into the Early Jurassic. Nonetheless, this is somewhat circular reasoning because no radiometric dates bind these sections over and above the biostratigraphy.

Lastly, the Early Jurassic Kayenta Formation magnetostratigraphy (Steiner and Tanner, 2014) is dominated by a long reverse polarity period which is punctuated by several short normal polarity intervals. This magnetic polarity stratigraphy matches the Early Pliensbachian (e-Pli R) polarity zone, for which a range of ~187–190 Ma has been provided (Steiner and Tanner, 2014). The Jurassic, globally, is typified by a fairly rapid and constant rate of polarity inversion (McElhinny and McFadden, 2000). These are not replicated in the Elliot Formation magnetostratigraphic composite section.

4.3. Magnetostratigraphic constraints

4.3.1. Problem 1: biased overprint by the Pliensbachian Karoo LIP

Given that the present day magnetic field is very similar to that of the Early Jurassic, and that the Karoo magmatism is Early Jurassic in age, overprinting associated with the Pliensbachian Karoo magmatism cannot be dismissed (e.g., Beaufort Group rocks, Kirschvink and Ward, 1998). However, other studies in the main Karoo basin show that the thermal effects are limited to rock close to intrusions (e.g., Maré et al., 2014; Smithard et al., 2015). Furthermore, chemical remanent magnetisation suggested by De Kock and Kirschvink (2004) provides low-temperature present field overprinting through the oxidation of magnetite to maghemite during weathering. Thus, given the (i) relatively well-defined, high temperature reverse polarity magnetization and (ii) the relatively high (>350 °C) laboratory unblocking temperatures of normal polarity magnetization, it is likely that the results presented here are not expressly exaggerated by any Early Jurassic normal overprinting. This is consistent with other palaeomagnetic studies throughout the main Karoo Basin (e.g., Lanci et al., 2013; Maré et al., 2014; Gastaldo et al., 2015), where primary (early acquired) normal polarity magnetisations were repeatedly isolated at temperatures >450 °C.

Samples from the Elliot Formation strata that pass the reversals test are more likely to indicate less bias by the younger overprint. Conversely, those that exhibit failures of the test (e.g., the UEF and combined Elliot Formation at Barkly Pass) indicate that the normal polarity samples may be biased by a younger normal overprint of either the Pliensbachian Karoo igneous event or of the present field (or both). Care has thus been taken in the interpretations carried forward into

the magnetic polarity stratigraphy. We emphasise that attempting to distinguish and separate out the cause of the bias is not possible given the similarity of (and overlap between) the normal Earliest Jurassic polarity to that of the Pliensbachian KLIP. Subsequently, all normal polarity magnetization results presented here, given the above discussion, have been ruled out as being complete (or near complete) overprints due to Pliensbachian Karoo magmatism.

4.3.2. Problem 2: continental sedimentation vs. magnetostratigraphy of the Elliot Formation

The magnetostratigraphy of continental red beds, which are typified by sedimentation regimes with vacillating rates of accumulation and syn-sedimentary erosion, as well as stratigraphic gaps of unknown magnitude and frequency (Miall, 2014a, 2014b), are strictly not comparable to that of stratigraphically more complete deeper lacustrine or marine depositional units. The fluctuations in sedimentation rate and the likelihood of stratigraphic gaps that resulted from abrupt events during the deposition of the Elliot Formation are limiting factors for the temporal resolution of the newly established magnetochrons of this unit.

From the sedimentological view point, the UEF, in particular, is more likely to represent a partially preserved sedimentary record with a complex history of low and continuous deposition as well as flash floods with rapid syn-sedimentary scouring and deposition (i.e., stationary vs. non-stationary episodes). The upper part of the Elliot Formation contains multiple palaeosol horizons and flash flood-type deposits, which suggest spatiotemporally variable preservation rates. A review by Marriott and Wright (2006) of the polygenetic origin of palaeosols shows that ancient soils can result from slow aggradational building of the soil profile followed by rapid scouring and truncation. The cyclic nature of these landscape evolutionary processes can repeatedly reactivate different soil horizons, and in particular in semi-arid settings, can result in reduced sections that embrace long stratigraphic intervals (e.g., Kraus, 1999; Gastaldo and Demko, 2011; Miall, 2014a, 2014b). In contrast, the LEF is likely to represent more continuous deposition, with fewer stratigraphic gaps relative to the UEF. Sampling was therefore adjusted (see Materials and methods) for pedogenically altered versus pedogenically unaltered mudstones.

5. Conclusion

The Upper Triassic to Lower Jurassic Elliot Formation in the main Karoo Basin is generally thought to represent a geologic record of almost 20 million years in the southern Gondwana, and thus it provides a case study for the complexities of magnetostratigraphic correlation of incomplete red bed sequences that are typical in continental settings.

Notwithstanding the inherent challenges, the magnetostratigraphic results of this study allowed the refinement of the age range of the Elliot Formation to between ~214-213 Ma and ~195–~190 Ma. Although, a maximim age for the lowermost LEF of ~ 218/ 219 Ma cannot be ruled out. The LEF-UEF boundary lies at the base of a normal magnetic polarity chron EF6.2n and above the EF6.1r. The latter may be coincident with the ETE but further refinements in biostratigraphy and radiometric ages would aid this hypothesis. The TJB is likely to be within the EF6.2n or within EF7n, based on comparison with the works of Hüsing et al. (2014). Thus, the results show that the position of the TJB is likely to be within the UEF and not at the LEF-UEF boundary. If valid, this can account for the traditional Triassic fauna found within the lower UEF (e.g., Moyeni tracksite chirotheroid-type trackways; Smith et al., 2009). The UEF is otherwise strongly correlated with Lower Jurassic magnetostratigraphy and fossil fauna (where available) of North America, which place it into the Hettangian and Early Sinemurian. The latter age is consistent with various lines of evidence (e.g., the conformably overlying Drakensberg Group lavas are ~183 Ma old).

The magnetostratigraphic approach used in this study, however, has inherent problems associated with: 1) efficacy of magnetic polarity stratigraphy and correlation of continental successions (e.g., unquantified sediment preservation rates, degree of influence of Pliensbachian Karoo dolerite intrusions); and 2) biostratigraphy of the Elliot Formation. The latter is rooted not only in the punctuated rock record of the unit (i.e., stratigraphic gaps of unknown duration and frequency) but also in the poorly defined biostratigraphic range zones of its fossil vertebrates. Addressing the latter requires, most of all, undertaking the long overdue revision of the taxonomic status of its dinosaur fauna. In summary, to further establish the stratigraphic value of this Upper Triassic to Lower Jurassic continental red bed succession for understanding the ETE event, including the evolutionary rates of the vertebrate fauna, and to increase its utility for global stratigraphic correlations, an urgent and systematic absolute dating of the Elliot Formation is needed. This could be achieved only by building on its existing litho-, bio- and magnetostratigraphic frameworks.

Grant disclosures

The following grant information was disclosed by the authors:
African Origins Programme of the National Research Foundation (NRF) of South Africa.
NRF/DAAD doctoral scholarship.
DST-NRF Centre of Excellence in Palaeosciences.
UCT Postgraduate Publication Incentive (PPI) award.

Acknowledgments

We would like thank the Basotho people for access to their communal land and the Lesotho Department of Mines and Geology for permitting (NR/M/E/10). LS would like to thank the South African National Parks (Golden Gate National Park; SCICL1146) for allowing sampling to be conducted on SANPark land. Many thanks for repeated drilling support during field work by colleagues William Krummeck and Adam Moodley. Research funds received from African Origins Programme of the National Research Foundation (Grant no. 77782 and 82606) of South Africa (by EB as principal investigator; MdK, FK as co-investigators) are gratefully acknowledged. LS would also like to acknowledge the support of the NRF/DAAD doctoral scholarship, and DST-NRF Centre of Excellence in Palaeosciences (CoE in Palaeosciences) postdoctoral fellowship and the UCT Postgraduate Publication Incentive (PPI) award. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript. Opinions expressed and conclusions arrived at are those of the authors and are not necessarily to be attributed to the funders. We are also very grateful to the Editor, Joseph Meert, as well as the generous comments of John Geissman and an anonymous reviewer for assisting in greatly improving this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.gr.2017.07.009.

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